

Vector Calculus, Coordinate Systems, Tensor Operations, and Complex Numbers

Modular Formula Completion and Practice Workbook

OT Science Notes

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How to Use the Modular Workbook

This combined file imports each modular part. Each numbered file can also be compiled and printed independently. The design keeps formulas in formula boxes, places practice work in one larger box per major section, and leaves ordinary explanation as normal text so that the page does not become visually noisy.

When a box uses a `..boxnosplit` environment, it is because the material should remain together on one page where possible. Longer practice boxes are allowed to break across pages.

1 Gradient, Divergence and Curl of Vector Fields

This part fixes the central grammar of vector calculus. The same symbol ∇ behaves differently depending on what it acts on and how it is combined with dot, cross, or ordinary multiplication. The fastest way to avoid confusion is to always ask what kind of object must come out of the operation.

Operation	Input	Output
∇f	scalar field	vector field
$\nabla \cdot \mathbf{A}$	vector field	scalar field
$\nabla \times \mathbf{A}$	vector field	vector field
$\nabla \mathbf{A}$	vector field	second-order tensor/Jacobian
$\nabla^2 f$	scalar field	scalar field
$\nabla^2 \mathbf{A}$	vector field	vector field, component by component

1.1 The Cartesian Nabla Operator

In rectangular Cartesian coordinates, the basis vectors $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ are constant. This is why the Cartesian formulas look cleaner than cylindrical or spherical formulas. The derivatives act only on the components of the fields, not on the basis vectors.

Core Cartesian Formulae

For a scalar field $f(x, y, z)$ and a vector field

$$\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}},$$

$$\nabla = \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z}.$$

$$\nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{i}} + \frac{\partial f}{\partial y} \hat{\mathbf{j}} + \frac{\partial f}{\partial z} \hat{\mathbf{k}}.$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}.$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{\mathbf{i}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{\mathbf{j}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{\mathbf{k}}.$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$

1.2 Gradient of a Scalar Field

The gradient points in the direction of steepest increase of a scalar field. Its magnitude is the maximum possible directional rate of change. If f is temperature, then ∇f points toward the direction in which temperature increases fastest.

Worked Example: Gradient of a Scalar Field

Let

$$f = x^2 y + y z^3.$$

Then

$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial f}{\partial y} = x^2 + z^3, \quad \frac{\partial f}{\partial z} = 3yz^2.$$

Therefore

$$\nabla f = 2xy \hat{\mathbf{i}} + (x^2 + z^3) \hat{\mathbf{j}} + 3yz^2 \hat{\mathbf{k}}.$$

1.3 Gradient of a Vector Field

The expression $\nabla \mathbf{A}$ is not the same kind of object as ∇f . The gradient of a scalar is a vector, but the gradient of a vector is a second-order tensor. It tells us how every component of $\mathbf{A}$ changes in every coordinate direction.

Using the convention

$$(\nabla \mathbf{A})_{ij} = \frac{\partial A_i}{\partial x_j},$$

the vector gradient is the matrix

$$\nabla \mathbf{A} = \begin{pmatrix} \partial_x A_x & \partial_y A_x & \partial_z A_x \\ \partial_x A_y & \partial_y A_y & \partial_z A_y \\ \partial_x A_z & \partial_y A_z & \partial_z A_z \end{pmatrix}.$$

The divergence is the trace of this tensor:

$$\nabla \cdot \mathbf{A} = \text{tr}(\nabla \mathbf{A}).$$

The curl is related to the skew-symmetric part of $\nabla \mathbf{A}$. This is why curl is naturally connected to rotation.

Vector Gradient, Divergence and Curl Relationship

$$(\nabla \mathbf{A})_{ij} = \partial_j A_i.$$

$$\nabla \cdot \mathbf{A} = \partial_i A_i = \text{tr}(\nabla \mathbf{A}).$$

$$(\nabla \times \mathbf{A})_i = \varepsilon_{ijk} \partial_j A_k.$$

If

$$W_{ij} = \frac{1}{2}(\partial_j A_i - \partial_i A_j),$$

then W_{ij} is the skew part of the vector gradient. In three dimensions, the curl is the axial vector associated with this skew part.

1.4 Divergence

Divergence measures source-like or sink-like behaviour. A positive divergence at a point suggests local outward spreading. A negative divergence suggests local inward concentration. Zero divergence does not mean the vector field is zero; it means the field has no net local source or sink.

1.5 Curl

Curl measures local rotational tendency. A non-zero curl indicates that a small paddle wheel placed in the field would tend to rotate. A zero curl means the field is irrotational locally, although this does not automatically settle global questions in domains with holes.

1.6 Directional Derivatives

For a scalar field f , the directional derivative in a unit direction $\hat{\mathbf{u}}$ is

$$D_{\hat{\mathbf{u}}} f = \nabla f \cdot \hat{\mathbf{u}}.$$

For a vector field, the derivative along $\hat{\mathbf{u}}$ is

$$(\hat{\mathbf{u}} \cdot \nabla) \mathbf{A}.$$

This gives another vector, not a scalar.

Directional Derivative Formulae

If

$$\hat{\mathbf{u}} = u_x \hat{\mathbf{i}} + u_y \hat{\mathbf{j}} + u_z \hat{\mathbf{k}}, \quad \|\hat{\mathbf{u}}\| = 1,$$

then

$$D_{\hat{\mathbf{u}}} f = u_x \frac{\partial f}{\partial x} + u_y \frac{\partial f}{\partial y} + u_z \frac{\partial f}{\partial z}.$$

$$\max_{\|\hat{\mathbf{u}}\|=1} D_{\hat{\mathbf{u}}} f = \|\nabla f\|, \quad \text{direction of maximum increase} = \frac{\nabla f}{\|\nabla f\|}.$$

$$(\hat{\mathbf{u}} \cdot \nabla) \mathbf{A} = (u_x \partial_x + u_y \partial_y + u_z \partial_z) \mathbf{A}.$$

1.7 Product and Composition Identities

These identities are worth learning in families. Notice how scalar multiplication, dot products and cross products change the type of the output.

Core Vector Identities

$$\nabla(fg) = f\nabla g + g\nabla f.$$

$$\nabla \cdot (f\mathbf{A}) = \nabla f \cdot \mathbf{A} + f\nabla \cdot \mathbf{A}.$$

$$\nabla \times (f\mathbf{A}) = \nabla f \times \mathbf{A} + f\nabla \times \mathbf{A}.$$

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} + \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}).$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}).$$

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}.$$

$$\nabla \times (\nabla f) = \mathbf{0}, \quad \nabla \cdot (\nabla \times \mathbf{A}) = 0.$$

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

Common Mistakes

1. Do not say that $\nabla \mathbf{A}$ is automatically a vector. It is usually a second-order tensor.
2. Do not confuse $\nabla \cdot \mathbf{A}$ with $\nabla \mathbf{A}$. Divergence is only the trace of the vector gradient in Cartesian coordinates.
3. Do not use the Cartesian curl formula unchanged in cylindrical or spherical coordinates.
4. Do not forget to normalise the direction vector before computing a directional derivative.
5. Do not treat ∇ as an ordinary vector in every context. It is a differential operator.

Quick Drills and Practice: Gradient, Divergence and Curl**Level A: Formula recall**

1. Complete: $\nabla = \hat{\mathbf{i}} \quad + \hat{\mathbf{j}} \quad + \hat{\mathbf{k}} \quad$.
2. Complete: $\nabla f = \quad$.
3. Complete: $\nabla \cdot \mathbf{A} = \quad$.
4. Complete: $(\nabla \times \mathbf{A})_x = \quad$.
5. Complete: $(\nabla \times \mathbf{A})_y = \quad$.
6. Complete: $(\nabla \times \mathbf{A})_z = \quad$.
7. Complete: $D_{\hat{\mathbf{u}}}f = \quad$.
8. Complete: $\nabla \times (\nabla f) = \quad$.
9. Complete: $\nabla \cdot (\nabla \times \mathbf{A}) = \quad$.
10. Complete: $\nabla \times (\nabla \times \mathbf{A}) = \quad$.

Level B: Direct computation

1. Compute ∇f for $f = x^2 + 3y^2 - z^2$.
2. Compute ∇f for $f = xy^2 + z \sin x$.
3. Compute $\nabla \cdot \mathbf{A}$ for $\mathbf{A} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$.
4. Compute $\nabla \cdot \mathbf{A}$ for $\mathbf{A} = yz\hat{\mathbf{i}} + xz\hat{\mathbf{j}} + xy\hat{\mathbf{k}}$.
5. Compute $\nabla \times \mathbf{A}$ for $\mathbf{A} = (-y)\hat{\mathbf{i}} + x\hat{\mathbf{j}} + 0\hat{\mathbf{k}}$.
6. Compute $\nabla \times \mathbf{A}$ for $\mathbf{A} = z\hat{\mathbf{i}} + x\hat{\mathbf{j}} + y\hat{\mathbf{k}}$.
7. For $f = x^2y + yz^2$, find $D_{\hat{\mathbf{u}}}f$ at $(1, 2, 1)$ in the direction $\mathbf{u} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$.
8. For $\mathbf{A} = x^2\hat{\mathbf{i}} + xy\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}$, compute $\nabla \mathbf{A}$, $\nabla \cdot \mathbf{A}$, and $\nabla \times \mathbf{A}$.

Level C: Interpretation and multi-step questions

1. A vector field has $\nabla \cdot \mathbf{A} > 0$ near a point. Explain the local behaviour in words.
2. Give a vector field with zero divergence but non-zero curl.
3. Give a vector field with non-zero divergence but zero curl.
4. If $\nabla f = \mathbf{0}$ throughout a connected region, what can be said about f ?
5. Find the direction of maximum increase of $f = x^2 + y^2 + z^2$ at $(1, 1, 2)$.
6. Find the maximum directional derivative of $f = xy + yz + zx$ at $(1, 2, 3)$.

Level D: Proofs and identities

1. Prove in Cartesian components that $\nabla \times (\nabla f) = \mathbf{0}$, assuming mixed partial derivatives are equal.
2. Prove in Cartesian components that $\nabla \cdot (\nabla \times \mathbf{A}) = 0$.
3. Verify the curl-curl identity for $\mathbf{A} = x^2\hat{\mathbf{i}} + y^2\hat{\mathbf{j}} + z^2\hat{\mathbf{k}}$.
4. Derive the product rule for $\nabla \cdot (f\mathbf{A})$.
5. Derive the product rule for $\nabla \times (f\mathbf{A})$.

Level E: Challenge applications

1. In fluid mechanics, $\mathbf{v}$ is a velocity field. Interpret $\nabla \cdot \mathbf{v} = 0$.
2. In electromagnetism, $\nabla \times \mathbf{E} = \mathbf{0}$ in electrostatics. What does this imply about a scalar potential?
3. For $\mathbf{v} = (-\omega y)\hat{\mathbf{i}} + (\omega x)\hat{\mathbf{j}}$, compute divergence and curl, then interpret the motion.
4. For $\mathbf{v} = ax\hat{\mathbf{i}} + ay\hat{\mathbf{j}} + az\hat{\mathbf{k}}$, compute divergence and curl, then interpret the motion.

2 Coordinate Systems and Tensor Operations

Coordinate systems are not just different labels for the same point. They also change how basis vectors, scale factors, area elements, volume elements and differential operators are written. Tensor operations help us keep the geometry clear when the coordinates become less simple than rectangular coordinates.

2.1 Rectangular, Cylindrical and Spherical Coordinates

Rectangular coordinates are the easiest because the basis vectors are fixed. Cylindrical and spherical coordinates are more natural for circular, axial, radial and angular symmetry, but their basis vectors change with position.

Coordinate Transformations

Cartesian: (x, y, z) .

Cylindrical: $x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z.$

$$\rho = \sqrt{x^2 + y^2}, \quad \phi = \text{atan2}(y, x), \quad z = z.$$

Spherical: $x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta.$

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \cos^{-1} \left(\frac{z}{r} \right), \quad \phi = \text{atan2}(y, x).$$

$$\rho = r \sin \theta, \quad z = r \cos \theta, \quad r = \sqrt{\rho^2 + z^2}.$$

2.2 Basis Vectors and Scale Factors

A coordinate basis is obtained by differentiating the position vector with respect to the coordinates. In orthogonal coordinates, the scale factors tell us how much physical length corresponds to a small coordinate change.

Basis Vectors, Scale Factors and Metric Matrices

For orthogonal coordinates (q^1, q^2, q^3) ,

$$\mathbf{g}_i = \frac{\partial \mathbf{r}}{\partial q^i}, \quad h_i = \|\mathbf{g}_i\|, \quad \hat{\mathbf{e}}_i = \frac{\mathbf{g}_i}{h_i}.$$

$$ds^2 = h_1^2 (dq^1)^2 + h_2^2 (dq^2)^2 + h_3^2 (dq^3)^2.$$

$$[g_{ij}] = \begin{pmatrix} h_1^2 & 0 & 0 \\ 0 & h_2^2 & 0 \\ 0 & 0 & h_3^2 \end{pmatrix}, \quad [g^{ij}] = \begin{pmatrix} 1/h_1^2 & 0 & 0 \\ 0 & 1/h_2^2 & 0 \\ 0 & 0 & 1/h_3^2 \end{pmatrix}.$$

Cartesian: $h_x = h_y = h_z = 1, \quad ds^2 = dx^2 + dy^2 + dz^2.$

Cylindrical: $h_\rho = 1, \quad h_\phi = \rho, \quad h_z = 1, \quad ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2.$

Spherical: $h_r = 1, \quad h_\theta = r, \quad h_\phi = r \sin \theta,$
 $ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$

2.3 Volume and Area Elements

The volume element is controlled by the product of the scale factors. This is why cylindrical and spherical integrals contain the extra factors ρ and $r^2 \sin \theta$.

Differential Elements

$$dV = h_1 h_2 h_3 dq^1 dq^2 dq^3.$$

$$dV_{\text{cart}} = dx dy dz.$$

$$dV_{\text{cyl}} = \rho d\rho d\phi dz.$$

$$dV_{\text{sph}} = r^2 \sin \theta dr d\theta d\phi.$$

For orthogonal coordinates, the coordinate surface elements are

$$dS_1 = h_2 h_3 dq^2 dq^3, \quad dS_2 = h_3 h_1 dq^3 dq^1, \quad dS_3 = h_1 h_2 dq^1 dq^2.$$

2.4 Coordinate Basis, Physical Components and Tensor Components

A vector can be represented in coordinate basis vectors or in unit basis vectors. In many engineering texts, cylindrical and spherical vector fields are written using unit basis vectors, so the listed components are physical components. In tensor notation, however, components may be covariant or contravariant depending on the basis used.

Basic Tensor Operations

For a second-order tensor A_{ij} ,

$$(\mathbf{A}\mathbf{x})_i = A_{ij}x_j, \quad \text{tr}(\mathbf{A}) = A_{ii}, \quad \mathbf{A} : \mathbf{B} = A_{ij}B_{ij}.$$

$$(\mathbf{AB})_{ij} = A_{ik}B_{kj}.$$

$$\mathbf{x} \cdot \mathbf{Ax} = x_i A_{ij} x_j.$$

$$\nabla \mathbf{A} \quad \text{for a vector field gives} \quad (\nabla \mathbf{A})_{ij} = \partial_j A_i.$$

$$\nabla \cdot \mathbf{T} \quad \text{for a second-order tensor can be written} \quad (\nabla \cdot \mathbf{T})_i = \partial_j T_{ij}.$$

2.5 Gradient, Divergence and Curl in Orthogonal Coordinates

The following formulas use physical components A_1, A_2, A_3 along unit vectors $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3$.

General Orthogonal Differential Operators

$$\nabla f = \sum_{i=1}^3 \hat{\mathbf{e}}_i \frac{1}{h_i} \frac{\partial f}{\partial q^i}.$$

$$\nabla \cdot \mathbf{A} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q^1} (h_2 h_3 A_1) + \frac{\partial}{\partial q^2} (h_3 h_1 A_2) + \frac{\partial}{\partial q^3} (h_1 h_2 A_3) \right].$$

$$\nabla \times \mathbf{A} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{\mathbf{e}}_1 & h_2 \hat{\mathbf{e}}_2 & h_3 \hat{\mathbf{e}}_3 \\ \partial_{q^1} & \partial_{q^2} & \partial_{q^3} \\ h_1 A_1 & h_2 A_2 & h_3 A_3 \end{vmatrix}.$$

2.6 Cylindrical Differential Operators

Cylindrical: (ρ, ϕ, z)

$$\nabla f = \hat{\mathbf{e}}_\rho \frac{\partial f}{\partial \rho} + \hat{\phi} \frac{1}{\rho} \frac{\partial f}{\partial \phi} + \hat{\mathbf{k}} \frac{\partial f}{\partial z}.$$

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}.$$

$$\nabla \times \mathbf{A} = \hat{\mathbf{e}}_\rho \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \hat{\phi} \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) + \hat{\mathbf{k}} \left(\frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\phi) - \frac{1}{\rho} \frac{\partial A_\rho}{\partial \phi} \right).$$

2.7 Spherical Differential Operators

Spherical: (r, θ, ϕ)

$$\nabla f = \hat{\mathbf{e}}_r \frac{\partial f}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r} \frac{\partial f}{\partial \theta} + \hat{\mathbf{e}}_\phi \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi}.$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta}(\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}.$$

$$\begin{aligned} \nabla \times \mathbf{A} = & \hat{\mathbf{e}}_r \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta}(\sin \theta A_\phi) - \frac{\partial A_\theta}{\partial \phi} \right] \\ & + \hat{\mathbf{e}}_\theta \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r}(r A_\phi) \right] + \hat{\mathbf{e}}_\phi \frac{1}{r} \left[\frac{\partial}{\partial r}(r A_\theta) - \frac{\partial A_r}{\partial \theta} \right]. \end{aligned}$$

Quick Drills and Practice: Coordinates and Tensor Operations

Level A: Coordinate conversion

- Convert $(x, y, z) = (3, 4, 5)$ to cylindrical coordinates.
- Convert $(x, y, z) = (1, 1, \sqrt{2})$ to spherical coordinates.
- Convert $(\rho, \phi, z) = (2, \pi/3, 5)$ to Cartesian coordinates.
- Convert $(r, \theta, \phi) = (4, \pi/2, \pi/6)$ to Cartesian coordinates.
- Fill in: $\rho = \underline{\hspace{2cm}}, z = \underline{\hspace{2cm}}, r = \underline{\hspace{2cm}}.$
- Fill in: $dV_{\text{cyl}} = \underline{\hspace{2cm}}.$
- Fill in: $dV_{\text{sph}} = \underline{\hspace{2cm}}.$
- Fill in the spherical scale factors.

Level B: Scale factors and metric matrices

- Starting from $\mathbf{r} = \rho \cos \phi \hat{\mathbf{i}} + \rho \sin \phi \hat{\mathbf{j}} + z \hat{\mathbf{k}}$, derive h_ρ, h_ϕ, h_z .
- Starting from $\mathbf{r} = r \sin \theta \cos \phi \hat{\mathbf{i}} + r \sin \theta \sin \phi \hat{\mathbf{j}} + r \cos \theta \hat{\mathbf{k}}$, derive h_r, h_θ, h_ϕ .
- Write the metric matrix $[g_{ij}]$ for cylindrical coordinates.
- Write the metric matrix $[g_{ij}]$ for spherical coordinates.
- Find ds^2 for cylindrical coordinates and explain why the term $\rho^2 d\phi^2$ appears.
- Find ds^2 for spherical coordinates and explain the term $r^2 \sin^2 \theta d\phi^2$.

Level C: Differential operators

- Compute ∇f in cylindrical coordinates for $f = \rho^2 z \sin \phi$.
- Compute ∇f in spherical coordinates for $f = r^2 \cos \theta$.
- Compute $\nabla \cdot \mathbf{A}$ in cylindrical coordinates for $\mathbf{A} = \rho^2 \hat{\mathbf{e}}_\rho + \rho z \hat{\phi} + z^2 \hat{\mathbf{k}}$.
- Compute $\nabla \cdot \mathbf{A}$ in spherical coordinates for $\mathbf{A} = r^2 \hat{\mathbf{e}}_r + r \sin \theta \hat{\mathbf{e}}_\theta + 0 \hat{\mathbf{e}}_\phi$.
- Compute $\nabla \times \mathbf{A}$ in cylindrical coordinates for $\mathbf{A} = 0 \hat{\mathbf{e}}_\rho + \rho^2 \hat{\phi} + 0 \hat{\mathbf{k}}$.
- Compute $\nabla \times \mathbf{A}$ in spherical coordinates for $\mathbf{A} = 0 \hat{\mathbf{e}}_r + 0 \hat{\mathbf{e}}_\theta + r \sin \theta \hat{\mathbf{e}}_\phi$.

Level D: Tensor operations

1. Given $A_{ij} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 4 & 0 & 2 \end{pmatrix}$, compute $A_{ij}x_j$ for $\mathbf{x} = (1, 2, 3)$.
2. Compute $\text{tr}(A)$, $A : A$, and A^2 for the matrix above.
3. If $T_{ij} = x_i x_j$, compute $\partial_j T_{ij}$ in three-dimensional Cartesian coordinates.
4. For $\mathbf{A} = x^2 \hat{\mathbf{i}} + xy \hat{\mathbf{j}} + yz \hat{\mathbf{k}}$, compute $(\nabla \mathbf{A})_{ij}$.
5. Show explicitly that $\nabla \cdot \mathbf{A} = \text{tr}(\nabla \mathbf{A})$ for the same vector field.

3 Index Notation, Kronecker Delta and Levi-Civita Symbol

Index notation compresses vector and tensor formulas into short expressions. It is especially useful when proving vector identities. The price is that every index must be handled carefully.

3.1 Einstein Summation Convention

When an index is repeated once as a subscript or repeated in a product in Euclidean notation, it is summed over its full range. In three-dimensional Cartesian coordinates, the range is usually 1, 2, 3.

Einstein Summation

$$a_i b_i = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

$$A_{ij} b_j = A_{i1} b_1 + A_{i2} b_2 + A_{i3} b_3.$$

$$A_{ij} B_{ij} = \sum_{i=1}^3 \sum_{j=1}^3 A_{ij} B_{ij}.$$

$$(AB)_{ij} = A_{ik} B_{kj}.$$

A free index appears on both sides of an equation and is not summed. A dummy index is repeated and summed. Dummy indices can be renamed as long as no conflict is created.

3.2 Kronecker Delta

The Kronecker delta is the identity tensor in component form. It replaces one index by another.

Kronecker Delta Identities

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

$$\delta_{ij} a_j = a_i, \quad \delta_{ij} a_i = a_j.$$

$$\delta_{ij} \delta_{jk} = \delta_{ik}, \quad \delta_{ii} = 3.$$

$$\delta_{ij}A_{ij} = A_{ii} = \text{tr}(A).$$

$$\partial_i x_j = \frac{\partial x_j}{\partial x_i} = \delta_{ij}.$$

$$\partial_i x_i = 3, \quad \partial_i r = \frac{x_i}{r}, \quad \partial_i (r^2) = 2x_i.$$

3.3 Levi-Civita Symbol

The Levi-Civita symbol tracks orientation and cross products. It is zero whenever any two indices are repeated.

Levi-Civita Identities

$$\varepsilon_{ijk} = \begin{cases} +1, & (i, j, k) \text{ is an even permutation of } (1, 2, 3), \\ -1, & (i, j, k) \text{ is an odd permutation of } (1, 2, 3), \\ 0, & \text{two indices are repeated.} \end{cases}$$

$$(\mathbf{a} \times \mathbf{b})_i = \varepsilon_{ijk} a_j b_k.$$

$$\varepsilon_{ijk} \varepsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}.$$

$$\varepsilon_{ijk} \varepsilon_{ijm} = 2\delta_{km}, \quad \varepsilon_{ijk} \varepsilon_{ijk} = 6.$$

$$\varepsilon_{ijk} \varepsilon_{lmn} = \begin{vmatrix} \delta_{il} & \delta_{im} & \delta_{in} \\ \delta_{jl} & \delta_{jm} & \delta_{jn} \\ \delta_{kl} & \delta_{km} & \delta_{kn} \end{vmatrix}.$$

3.4 Vector Calculus in Index Notation

Vector Operations in Index Form

$$(\nabla f)_i = \partial_i f.$$

$$\nabla \cdot \mathbf{A} = \partial_i A_i.$$

$$(\nabla \times \mathbf{A})_i = \varepsilon_{ijk} \partial_j A_k.$$

$$\nabla^2 f = \partial_i \partial_i f.$$

$$(\nabla \mathbf{A})_{ij} = \partial_j A_i.$$

$$(\nabla \cdot \mathbf{T})_i = \partial_j T_{ij}.$$

3.5 Proof Pattern: Curl of Curl

The index method is powerful because the proof becomes a controlled substitution. Starting with

$$(\nabla \times (\nabla \times \mathbf{A}))_i = \varepsilon_{ijk} \partial_j (\nabla \times \mathbf{A})_k,$$

use

$$(\nabla \times \mathbf{A})_k = \varepsilon_{klm} \partial_l A_m.$$

Then

$$(\nabla \times (\nabla \times \mathbf{A}))_i = \varepsilon_{ijk} \varepsilon_{klm} \partial_j \partial_l A_m.$$

Using the contraction identity gives the familiar result

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

Index-Notation Mistakes

1. Do not repeat the same index more than twice in one term unless you have explicitly defined what it means.
2. Do not change a free index on one side of an equation without changing it on the other side.
3. Dummy indices may be renamed, but free indices may not.
4. The identity $\varepsilon_{ijk} \varepsilon_{ilm}$ depends on which index is contracted. Check the repeated index carefully.

Quick Drills and Practice: Index Notation

Level A: Expansion and simplification

1. Expand $a_i b_i$ in three dimensions.
2. Expand $A_{ij} b_j$ for $i = 1$.
3. Simplify $\delta_{ij} a_j$.
4. Simplify $\delta_{ij} A_{jk}$.
5. Simplify $\delta_{ij} \delta_{jk} \delta_{kl}$.
6. Simplify $\delta_{ij} A_{ij}$.
7. Compute δ_{ii} in three dimensions.
8. Compute $\partial_i x_j$.
9. Compute $\partial_i x_i$.
10. Compute $\partial_i (x_i x_j)$.

Level B: Levi-Civita

1. Evaluate ε_{123} , ε_{132} , ε_{221} , ε_{312} .
2. Expand $(\mathbf{a} \times \mathbf{b})_1 = \varepsilon_{1jk} a_j b_k$.
3. Show that $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$ using ε_{ijk} .
4. Simplify $\varepsilon_{ijk} \varepsilon_{ijm}$.
5. Simplify $\varepsilon_{ijk} \varepsilon_{ijk}$.
6. Use $\varepsilon_{ijk} \varepsilon_{ilm}$ to simplify $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d})$.

Level C: Vector identities

1. Prove $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ using index notation.
2. Prove $\nabla \times (\nabla f) = \mathbf{0}$ using index notation.
3. Prove $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$.

4. Prove $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$.
5. Prove $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$.

Level D: Derivative identities

1. Show that $\partial_i r = x_i/r$, where $r = (x_j x_j)^{1/2}$.
2. Show that $\partial_i (1/r) = -x_i/r^3$.
3. Compute $\partial_i (x_i/r^3)$ for $r \neq 0$.
4. Compute $\partial_j (x_i x_j)$ in three dimensions.
5. If $T_{ij} = f \delta_{ij}$, compute $\partial_j T_{ij}$.

4 Covariant and Contravariant Bases, Metrics and General Curvilinear Calculus

This part connects coordinate geometry with tensor notation. The key idea is that a vector itself is geometric, but its components depend on the basis used to describe it. Covariant and contravariant components are two related ways of recording the same geometric object.

4.1 Coordinate Basis and Reciprocal Basis

Let

$$\mathbf{r} = \mathbf{r}(q^1, q^2, q^3).$$

The covariant basis vectors are tangent to coordinate curves:

$$\mathbf{g}_i = \frac{\partial \mathbf{r}}{\partial q^i}.$$

The contravariant basis vectors form the reciprocal basis:

$$\mathbf{g}^i \cdot \mathbf{g}_j = \delta_j^i.$$

Covariant and Contravariant Bases

$$\mathbf{g}_i = \frac{\partial \mathbf{r}}{\partial q^i}, \quad \mathbf{g}^i = \nabla q^i.$$

$$\mathbf{g}^i \cdot \mathbf{g}_j = \delta_j^i.$$

$$\mathbf{A} = A^i \mathbf{g}_i = A_i \mathbf{g}^i.$$

$$A^i = \mathbf{A} \cdot \mathbf{g}^i, \quad A_i = \mathbf{A} \cdot \mathbf{g}_i.$$

4.2 Metric Tensor

The metric tensor records dot products between basis vectors. It is the object that turns coordinate changes into physical lengths and angles.

Metric Tensor and Raising/Lowering

$$g_{ij} = \mathbf{g}_i \cdot \mathbf{g}_j, \quad g^{ij} = \mathbf{g}^i \cdot \mathbf{g}^j.$$

$$g^{ik} g_{kj} = \delta_j^i.$$

$$A_i = g_{ij} A^j, \quad A^i = g^{ij} A_j.$$

$$ds^2 = g_{ij} dq^i dq^j.$$

$$\mathbf{A} \cdot \mathbf{B} = g_{ij} A^i B^j = g^{ij} A_i B_j = A_i B^i = A^i B_i.$$

4.3 Orthogonal Coordinates as a Special Case

If the coordinate system is orthogonal, the metric tensor is diagonal. This is why scale factors are enough to describe many engineering coordinate systems.

Orthogonal Metric and Scale Factors

$$g_{ij} = h_i^2 \delta_{ij} \quad \text{no sum on } i.$$

$$g^{ij} = \frac{1}{h_i^2} \delta^{ij} \quad \text{no sum on } i.$$

$$\mathbf{g}_i = h_i \hat{\mathbf{e}}_i, \quad \mathbf{g}^i = \frac{1}{h_i} \hat{\mathbf{e}}_i.$$

$$A_i = h_i A_{\text{phys},i}, \quad A^i = \frac{A_{\text{phys},i}}{h_i}.$$

4.4 Jacobian, Metric and Volume

For a coordinate transformation $x^a = x^a(q^1, q^2, q^3)$, the Jacobian matrix connects small coordinate changes to small physical displacements.

Jacobian and Volume Element

$$J^a_i = \frac{\partial x^a}{\partial q^i}.$$

$$g_{ij} = \frac{\partial x^a}{\partial q^i} \frac{\partial x^a}{\partial q^j} = J^a_i J^a_j.$$

$$g = \det(g_{ij}), \quad \sqrt{g} = |\det(J)|.$$

$$dV = \sqrt{g} dq^1 dq^2 dq^3.$$

For orthogonal coordinates,

$$\sqrt{g} = h_1 h_2 h_3.$$

4.5 Gradient, Divergence and Laplacian in General Curvilinear Coordinates

The following formulas use the metric determinant $g = \det(g_{ij})$. They are especially useful when coordinates are not necessarily orthogonal.

General Curvilinear Formulae

$$\nabla f = (\partial_i f) \mathbf{g}^i.$$

$$\nabla \cdot \mathbf{A} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial q^i} (\sqrt{g} A^i).$$

$$\nabla^2 f = \frac{1}{\sqrt{g}} \frac{\partial}{\partial q^i} \left(\sqrt{g} g^{ij} \frac{\partial f}{\partial q^j} \right).$$

For orthogonal coordinates,

$$\nabla^2 f = \frac{1}{h_1 h_2 h_3} \sum_{i=1}^3 \frac{\partial}{\partial q^i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial f}{\partial q^i} \right).$$

4.6 Cylindrical and Spherical Laplacians

Common Laplacians

Cartesian:

$$\nabla^2 f = f_{xx} + f_{yy} + f_{zz}.$$

Cylindrical:

$$\nabla^2 f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}.$$

Spherical:

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}.$$

Quick Drills and Practice: Covariant/Contravariant Components and Metrics

Level A: Conceptual recall

- | | |
|---|---------------------------------------|
| 1. Complete: $\mathbf{g}_i =$ _____. | 5. Complete: $g^{ik} g_{kj} =$ _____. |
| 2. Complete: $\mathbf{g}^i =$ _____. | 6. Complete: $A_i =$ _____. |
| 3. Complete: $\mathbf{g}^i \cdot \mathbf{g}_j =$ _____. | 7. Complete: $A^i =$ _____. |
| 4. Complete: $g_{ij} =$ _____. | 8. Complete: $ds^2 =$ _____. |

Level B: Metric computations

1. Compute g_{ij} for cylindrical coordinates directly from $\mathbf{r}(\rho, \phi, z)$.
2. Compute g^{ij} for cylindrical coordinates.
3. Compute g_{ij} for spherical coordinates directly from $\mathbf{r}(r, \theta, \phi)$.
4. Compute g^{ij} for spherical coordinates.

5. Compute $\sqrt{g}$ for Cartesian, cylindrical and spherical coordinates.
6. Show that $dV = \sqrt{g} dq^1 dq^2 dq^3$ gives the correct cylindrical and spherical volume elements.

Level C: Components

1. In cylindrical coordinates, if the physical components are $(A_\rho, A_\phi, A_z) = (2, 3, 4)$, write the contravariant components A^i .
2. In spherical coordinates, if the physical components are $(A_r, A_\theta, A_\phi) = (1, 2, 3)$, write A^i and A_i .
3. Given $g_{ij} = \text{diag}(1, r^2, r^2 \sin^2 \theta)$, lower the index of $A^i = (r, \sin \theta, 1)$.
4. Given $g^{ij} = \text{diag}(1, 1/r^2, 1/(r^2 \sin^2 \theta))$, raise the index of $A_i = (r^2, r^2 \theta, r^2 \sin \theta)$.

Level D: Differential operators

1. Derive the cylindrical divergence formula from $\nabla \cdot \mathbf{A} = \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} A^i)$.
2. Derive the spherical divergence formula from $\nabla \cdot \mathbf{A} = \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} A^i)$.
3. Derive the cylindrical scalar Laplacian from the general curvilinear formula.
4. Derive the spherical scalar Laplacian from the general curvilinear formula.
5. Compute $\nabla^2 f$ in cylindrical coordinates for $f = \rho^2 + z^2$.
6. Compute $\nabla^2 f$ in spherical coordinates for $f = r^2$.

5 Symm and Skew-Symm Tensors, Invariants n' Decomps

Second-order tensors appear naturally as stress tensors, strain tensors, inertia tensors, conductivity tensors, deformation gradients and vector gradients. Their symmetric and skew-symmetric parts often carry different physical meanings.

5.1 Symmetric and Skew-Symmetric Tensors

A tensor is symmetric if transposing it does not change it. A tensor is skew-symmetric if transposing it changes its sign. In three dimensions, a skew-symmetric tensor has only three independent components and can be represented by an axial vector.

Symmetry Conditions

$$S_{ij} = S_{ji} \quad \text{symmetry.}$$

$$W_{ij} = -W_{ji} \quad \text{skew-symmetry.}$$

$$W_{ii} = 0.$$

$$A_{ij} = S_{ij} + W_{ij}.$$

$$S_{ij} = \frac{1}{2}(A_{ij} + A_{ji}), \quad W_{ij} = \frac{1}{2}(A_{ij} - A_{ji}).$$

5.2 Connection with Vector Gradient

If $\mathbf{v}$ is a velocity field, then $\nabla \mathbf{v}$ can be decomposed into symmetric and skew-symmetric parts. The symmetric part is related to strain rate, while the skew-symmetric part is related to local rotation.

Gradient Decomposition

$$L_{ij} = \partial_j v_i.$$

$$D_{ij} = \frac{1}{2}(L_{ij} + L_{ji}) \quad \text{symmetric rate-of-deformation tensor.}$$

$$\Omega_{ij} = \frac{1}{2}(L_{ij} - L_{ji}) \quad \text{skew spin tensor.}$$

$$L_{ij} = D_{ij} + \Omega_{ij}.$$

$$\omega_i = (\nabla \times \mathbf{v})_i = \varepsilon_{ijk} \partial_j v_k.$$

5.3 Principal Invariants of a Second-Order Tensor

Invariants do not change under a rotation of coordinates. They describe intrinsic properties of the tensor.

Principal Invariants

For a 3×3 tensor $\mathbf{A}$,

$$I_1 = \text{tr}(\mathbf{A}) = A_{ii}.$$

$$I_2 = \frac{1}{2} \left[(\text{tr} \mathbf{A})^2 - \text{tr}(\mathbf{A}^2) \right].$$

$$I_3 = \det(\mathbf{A}).$$

The characteristic polynomial may be written as

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \lambda^3 - I_1 \lambda^2 + I_2 \lambda - I_3.$$

Equivalently,

$$\det(\mathbf{A} - \lambda \mathbf{I}) = -\lambda^3 + I_1 \lambda^2 - I_2 \lambda + I_3.$$

5.4 Deviatoric and Spherical Parts

Many physical tensors can be decomposed into a mean or spherical part and a deviatoric part. In stress analysis, the spherical part is associated with hydrostatic pressure, while the deviatoric part is associated with shape distortion.

Deviatoric Decomposition

$$\text{sph}(A)_{ij} = \frac{1}{3} A_{kk} \delta_{ij}.$$

$$A'_{ij} = \text{dev}((A))_{ij} = A_{ij} - \frac{1}{3} A_{kk} \delta_{ij}.$$

$$A_{ij} = \frac{1}{3} A_{kk} \delta_{ij} + A'_{ij}.$$

$$A'_{kk} = 0.$$

For a stress tensor σ_{ij} ,

$$p = -\frac{1}{3} \sigma_{kk}, \quad s_{ij} = \sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij}.$$

5.5 Invariants of Deviatoric Tensors

For a deviatoric tensor A' , the first invariant is zero. The remaining invariants are important in mechanics.

Deviatoric Invariants

$$J_1 = \text{tr}(A') = 0.$$

$$J_2 = \frac{1}{2} A'_{ij} A'_{ij}.$$

$$J_3 = \det(A').$$

For stress deviator s_{ij} , one common convention is

$$J_2 = \frac{1}{2} s_{ij} s_{ij}.$$

Worked Example: Tensor Decomposition

Let

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 4 & 0 & -1 \\ 5 & 2 & 1 \end{pmatrix}.$$

Then

$$S = \frac{1}{2}(A + A^T), \quad W = \frac{1}{2}(A - A^T).$$

Compute

$$S = \begin{pmatrix} 2 & 5/2 & 4 \\ 5/2 & 0 & 1/2 \\ 4 & 1/2 & 1 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & -3/2 & -1 \\ 3/2 & 0 & -3/2 \\ 1 & 3/2 & 0 \end{pmatrix}.$$

Check that $S^T = S$, $W^T = -W$, and $A = S + W$.

Quick Drills and Practice: Tensor Decompositions and Invariants**Level A: Symmetry checks**

1. Is $\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ symmetric?
2. Is $\begin{pmatrix} 0 & -4 \\ 4 & 0 \end{pmatrix}$ skew-symmetric?
3. Complete: $S_{ij} - S_{ji} = \underline{\hspace{2cm}}$.
4. Complete: $W_{ij} + W_{ji} = \underline{\hspace{2cm}}$.
5. Complete: $W_{ii} = \underline{\hspace{2cm}}$.
6. Complete: $A_{ij} = \underline{\hspace{2cm}}$.
7. Complete: $S_{ij} = \underline{\hspace{2cm}}$.
8. Complete: $W_{ij} = \underline{\hspace{2cm}}$.

Level B: Decomposition

1. Decompose $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$ into symmetric and skew-symmetric parts.
2. Decompose $A = \begin{pmatrix} 0 & -1 & 2 \\ 1 & 0 & -3 \\ -2 & 3 & 0 \end{pmatrix}$. What do you notice?
3. For $\mathbf{v} = (-y, x, 0)$, compute $\nabla \mathbf{v}$, its symmetric part and its skew part.
4. For $\mathbf{v} = (ax, by, cz)$, compute $\nabla \mathbf{v}$, its trace and its symmetric/skew parts.

Level C: Invariants

1. Compute I_1, I_2, I_3 for $A = \text{diag}(1, 2, 3)$.
2. Compute I_1, I_2, I_3 for $A = \text{diag}(a, b, c)$.
3. Compute I_1, I_2, I_3 for $A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.
4. Write the characteristic polynomial of a 3×3 tensor using I_1, I_2, I_3 .
5. If a tensor has eigenvalues $\lambda_1, \lambda_2, \lambda_3$, express I_1, I_2, I_3 in terms of the eigenvalues.

Level D: Deviatoric tensors

1. Find the deviatoric part of $A = \text{diag}(4, 1, 1)$.
2. Find the deviatoric part of $A = \text{diag}(6, 3, 0)$.
3. Prove that $A'_{kk} = 0$.
4. For $A = \text{diag}(4, 1, 1)$, compute $J_2 = \frac{1}{2}A'_{ij}A'_{ij}$.
5. For $\sigma = \text{diag}(100, 50, 25)$, compute the hydrostatic part and deviatoric part.

Level E: Challenge

1. Show that the trace of a skew-symmetric tensor is zero.
2. Show that $x_i W_{ij} x_j = 0$ for any skew-symmetric tensor W_{ij} .
3. Show that $x_i S_{ij} x_j$ depends only on the symmetric part of a tensor.
4. If A is symmetric, explain why its eigenvalues are real in ordinary Euclidean space.

5. Explain why invariants are useful when comparing the same tensor in two different coordinate frames.

6 Complex Numbers, Circular Functions and Hyperbolic Functions

Complex numbers connect algebra, geometry, oscillations, rotations, exponentials and hyperbolic functions. The core bridge is Euler's formula. Once that is understood, trigonometric and hyperbolic identities become different faces of the same exponential structure.

6.1 Basic Complex Algebra

A complex number has a real part and an imaginary part:

$$z = x + iy, \quad i^2 = -1.$$

The complex plane treats x as the horizontal coordinate and y as the vertical coordinate.

Complex Number Basics

$$z = x + iy, \quad \operatorname{Re}(z) = x, \quad \operatorname{Im}(z) = y.$$

$$\bar{z} = x - iy.$$

$$|z| = \sqrt{x^2 + y^2}, \quad \arg(z) = \theta.$$

$$z\bar{z} = |z|^2.$$

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{x - iy}{x^2 + y^2}, \quad z \neq 0.$$

6.2 Polar and Exponential Forms

The polar form is often the best way to multiply, divide, raise to powers and extract roots.

Euler, Polar Form and De Moivre

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta}.$$

$$r = |z|, \quad \theta = \arg(z).$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}.$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}, \quad z_2 \neq 0.$$

$$(re^{i\theta})^n = r^n e^{in\theta}.$$

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta).$$

6.3 Roots of Complex Numbers

If $z^n = Re^{i\Theta}$, then the n roots are evenly spaced around a circle in the complex plane.

Complex Roots

$$z_k = R^{1/n} e^{i(\Theta+2\pi k)/n}, \quad k = 0, 1, 2, \dots, n-1.$$

Equivalently,

$$z_k = R^{1/n} \left[\cos \left(\frac{\Theta + 2\pi k}{n} \right) + i \sin \left(\frac{\Theta + 2\pi k}{n} \right) \right].$$

6.4 Complex Exponentials and Circular Functions

The ordinary circular functions can be written using complex exponentials:

Circular Functions from Exponentials

$$\cos x = \frac{e^{ix} + e^{-ix}}{2}.$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}.$$

$$\tan x = \frac{\sin x}{\cos x}.$$

$$\cos^2 x + \sin^2 x = 1.$$

$$e^{ix} e^{-ix} = 1.$$

6.5 Hyperbolic Functions

Hyperbolic functions are built from real exponentials. They resemble circular functions, but their signs differ in crucial places.

Hyperbolic Definitions and Identities

$$\sinh x = \frac{e^x - e^{-x}}{2}.$$

$$\cosh x = \frac{e^x + e^{-x}}{2}.$$

$$\tanh x = \frac{\sinh x}{\cosh x}.$$

$$\cosh^2 x - \sinh^2 x = 1.$$

$$\sinh(2x) = 2 \sinh x \cosh x.$$

$$\cosh(2x) = \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x.$$

$$\frac{d}{dx} \sinh x = \cosh x, \quad \frac{d}{dx} \cosh x = \sinh x, \quad \frac{d}{dx} \tanh x = \operatorname{sech}^2 x.$$

6.6 Circular—Hyperbolic Connections

Replacing a real argument by an imaginary one converts circular functions into hyperbolic functions and vice versa.

Complex Arguments and Hyperbolic Connections

$$\cos(ix) = \cosh x.$$

$$\sin(ix) = i \sinh x.$$

$$\tan(ix) = i \tanh x.$$

$$\cosh(ix) = \cos x.$$

$$\sinh(ix) = i \sin x.$$

$$\tanh(ix) = i \tan x.$$

6.7 Inverse Hyperbolic Functions

Inverse hyperbolic functions can be written using logarithms. These are important in integration and differential equations.

Inverse Hyperbolic Forms

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1}).$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}), \quad x \geq 1.$$

$$\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1.$$

Quick Drills and Practice: Complex and Hyperbolic Functions

Level A: Formula recall

- Complete: $e^{i\theta} = \underline{\hspace{2cm}}$.
- Complete: $z\bar{z} = \underline{\hspace{2cm}}$.
- Complete: $1/(x + iy) = \underline{\hspace{2cm}}$.
- Complete: $\cos x = \underline{\hspace{2cm}}$.
- Complete: $\sin x = \underline{\hspace{2cm}}$.
- Complete: $\sinh x = \underline{\hspace{2cm}}$.
- Complete: $\cosh x = \underline{\hspace{2cm}}$.
- Complete: $\cosh^2 x - \sinh^2 x = \underline{\hspace{2cm}}$.
- Complete: $\cos(ix) = \underline{\hspace{2cm}}$.
- Complete: $\sin(ix) = \underline{\hspace{2cm}}$.

Level B: Complex algebra

- Compute $(3 + 2i) + (1 - 5i)$.
- Compute $(2 + i)(3 - 4i)$.
- Compute $(1 + i)^4$ using ordinary algebra.
- Write $1 + i$ in polar and exponential form.
- Write $-1 + \sqrt{3}i$ in polar and exponential form.
- Compute $(1 + i)^8$ using polar form.
- Compute $\frac{2+3i}{1-i}$.

Level C: Powers and roots

- Find all cube roots of 8.
- Find all fourth roots of 1.
- Find all fourth roots of -16 .
- Solve $z^3 = -8i$.
- Sketch the roots of $z^5 = 1$ on the complex plane.

Level D: Hyperbolic connections

- Prove $\cos(ix) = \cosh x$ using exponential definitions.
- Prove $\sin(ix) = i \sinh x$.
- Prove $\cosh(ix) = \cos x$.
- Prove $\sinh(ix) = i \sin x$.
- Prove $\cosh^2 x - \sinh^2 x = 1$.
- Derive $\sinh(2x) = 2 \sinh x \cosh x$.
- Derive $\cosh(2x) = \cosh^2 x + \sinh^2 x$.

Level E: Challenge

- Express $\cos(x + iy)$ in terms of $\cos x, \sin x, \cosh y, \sinh y$.
- Express $\sin(x + iy)$ in terms of $\cos x, \sin x, \cosh y, \sinh y$.
- Solve $e^z = -1$ for complex z .
- Solve $\cos z = 0$ for complex z .
- Show that multiplication by $e^{i\theta}$ rotates a complex number by angle θ .

7 Final Mixed Practice Bank and Formula Recall

This final part is designed for spaced revision. It deliberately mixes recall, computation, proof, interpretation and engineering-style applications. Use it after completing the earlier parts, or use it diagnostically to find weak areas.

7.1 Formula Recall Page

Final Formula Recall

Fill these from memory before checking earlier sections.

- | | |
|--|---|
| 1. $\nabla f =$ _____ | 14. $A'_{ij} =$ _____ |
| 2. $\nabla \cdot \mathbf{A} =$ _____ | 15. $\mathbf{g}_i =$ _____ |
| 3. $\nabla \times \mathbf{A} =$ _____ | 16. $\mathbf{g}^i \cdot \mathbf{g}_j =$ _____ |
| 4. $D_{\hat{\mathbf{u}}} f =$ _____ | 17. $A_i =$ _____ |
| 5. $\nabla^2 f =$ _____ | 18. $A^i =$ _____ |
| 6. $(\nabla \mathbf{A})_{ij} =$ _____ | 19. $ds^2 =$ _____ |
| 7. $\delta_{ij} a_j =$ _____ | 20. $dV =$ _____ |
| 8. $\varepsilon_{ijk} \varepsilon_{ilm} =$ _____ | 21. $dV_{\text{cyl}} =$ _____ |
| 9. $S_{ij} =$ _____ | 22. $dV_{\text{sph}} =$ _____ |
| 10. $W_{ij} =$ _____ | 23. $e^{i\theta} =$ _____ |
| 11. $I_1 =$ _____ | 24. $\cos(ix) =$ _____ |
| 12. $I_2 =$ _____ | 25. $\sin(ix) =$ _____ |
| 13. $I_3 =$ _____ | 26. $\cosh^2 x - \sinh^2 x =$ _____ |

7.2 Part I: Vector Calculus Computation

Final Practice I: Grad, Div, Curl and Directional Derivatives

1. Compute ∇f for $f = x^3 y^2 + z \cos x$.
2. Compute ∇f for $f = \ln(x^2 + y^2 + z^2)$, away from the origin.
3. Compute $\nabla^2 f$ for $f = x^2 + y^2 + z^2$.
4. Compute $\nabla^2 f$ for $f = 1/r$, where $r = \sqrt{x^2 + y^2 + z^2}$, away from the origin.
5. Compute $\nabla \cdot \mathbf{A}$ for $\mathbf{A} = x^2 \hat{\mathbf{i}} + y^2 \hat{\mathbf{j}} + z^2 \hat{\mathbf{k}}$.
6. Compute $\nabla \cdot \mathbf{A}$ for $\mathbf{A} = yz \hat{\mathbf{i}} + xz \hat{\mathbf{j}} + xy \hat{\mathbf{k}}$.
7. Compute $\nabla \times \mathbf{A}$ for $\mathbf{A} = xy \hat{\mathbf{i}} + yz \hat{\mathbf{j}} + zx \hat{\mathbf{k}}$.
8. Compute $\nabla \times \mathbf{A}$ for $\mathbf{A} = (-y/(x^2 + y^2)) \hat{\mathbf{i}} + (x/(x^2 + y^2)) \hat{\mathbf{j}}$, away from the z -axis.
9. For $f = x^2 y + yz^2$, compute the directional derivative at $(1, 1, 2)$ in direction $(2, -1, 2)$.
10. Find the direction of steepest increase of $f = xy + yz + zx$ at $(1, 2, 3)$.

11. Compute $\nabla \mathbf{A}$ for $\mathbf{A} = x^2 \hat{\mathbf{i}} + xy \hat{\mathbf{j}} + xz \hat{\mathbf{k}}$.
12. For the vector field in Question 11, find the symmetric and skew-symmetric parts of $\nabla \mathbf{A}$.
13. Verify $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ for $\mathbf{A} = z \hat{\mathbf{i}} + x \hat{\mathbf{j}} + y \hat{\mathbf{k}}$.
14. Verify $\nabla \times (\nabla f) = 0$ for $f = xyz + x^2 + y^2$.
15. Evaluate $\nabla \times (\nabla \times \mathbf{A})$ directly and using the identity for $\mathbf{A} = x^2 \hat{\mathbf{i}} + y^2 \hat{\mathbf{j}} + z^2 \hat{\mathbf{k}}$.

7.3 Part II: Coordinate Systems and Differential Operators

Final Practice II: Coordinates, Scale Factors and Operators

1. Convert $(x, y, z) = (-1, \sqrt{3}, 2)$ to cylindrical coordinates using atan2.
2. Convert $(x, y, z) = (1, 1, \sqrt{2})$ to spherical coordinates.
3. Convert $(\rho, \phi, z) = (3, 5\pi/6, -2)$ to Cartesian coordinates.
4. Convert $(r, \theta, \phi) = (2, \pi/3, \pi/4)$ to Cartesian coordinates.
5. Derive the cylindrical basis vectors $\hat{\mathbf{e}}_\rho$ and $\hat{\boldsymbol{\phi}}$ in terms of $\hat{\mathbf{i}}, \hat{\mathbf{j}}$.
6. Derive the spherical basis vectors $\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi$ in terms of $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$.
7. Derive h_ρ, h_ϕ, h_z from the cylindrical position vector.
8. Derive h_r, h_θ, h_ϕ from the spherical position vector.
9. Write $[g_{ij}]$ and $[g^{ij}]$ for cylindrical coordinates.
10. Write $[g_{ij}]$ and $[g^{ij}]$ for spherical coordinates.
11. Compute ∇f in cylindrical coordinates for $f = \rho z \cos \phi$.
12. Compute ∇f in spherical coordinates for $f = r^3 \sin \theta \cos \phi$.
13. Compute $\nabla \cdot \mathbf{A}$ in cylindrical coordinates for $\mathbf{A} = \rho \hat{\mathbf{e}}_\rho + \rho^2 \hat{\boldsymbol{\phi}} + z \hat{\mathbf{k}}$.
14. Compute $\nabla \cdot \mathbf{A}$ in spherical coordinates for $\mathbf{A} = r \hat{\mathbf{e}}_r + \sin \theta \hat{\mathbf{e}}_\theta + \cos \phi \hat{\mathbf{e}}_\phi$.
15. Compute $\nabla \times \mathbf{A}$ in cylindrical coordinates for $\mathbf{A} = 0 \hat{\mathbf{e}}_\rho + \rho \hat{\boldsymbol{\phi}} + z \hat{\mathbf{k}}$.
16. Compute $\nabla \times \mathbf{A}$ in spherical coordinates for $\mathbf{A} = 0 \hat{\mathbf{e}}_r + 0 \hat{\mathbf{e}}_\theta + r \sin \theta \hat{\mathbf{e}}_\phi$.
17. Derive the cylindrical scalar Laplacian from the general orthogonal coordinate formula.
18. Derive the spherical scalar Laplacian from the general curvilinear formula.

7.4 Part III: Index Notation and Tensor Identities

Final Practice III: Index Notation, Delta and Levi-Civita

1. Expand $A_{ij}B_{jk}$ for the component $(i, k) = (1, 2)$.
2. Simplify $\delta_{ij}\delta_{jk}a_k$.
3. Simplify $\delta_{ij}A_{ik}B_{jk}$.
4. Simplify $\varepsilon_{ijk}\varepsilon_{ilm}a_jb_kc_ld_m$.
5. Show that $\varepsilon_{ijk}a_jb_k$ gives the cross product.
6. Prove $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$.
7. Prove $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.
8. Prove the vector triple product identity using ε_{ijk} .
9. Prove $\nabla \times (\nabla f) = 0$ using indices.
10. Prove $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ using indices.
11. Prove the curl-curl identity using indices.
12. Compute $\partial_i(x_jx_j)$.
13. Compute $\partial_j(x_ix_j)$ in three dimensions.
14. Compute $\partial_i(r^n)$, where $r = (x_jx_j)^{1/2}$.
15. Compute $\partial_i(x_i/r^3)$ for $r \neq 0$.

7.5 Part IV: Tensor Decompositions and Invariants

Final Practice IV: Symmetry, Skew-Symmetry and Invariants

1. Decompose $A = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 4 & 5 \\ 1 & -2 & 6 \end{pmatrix}$ into symmetric and skew parts.
2. Check whether $A = \begin{pmatrix} 0 & -2 & 1 \\ 2 & 0 & -3 \\ -1 & 3 & 0 \end{pmatrix}$ is skew-symmetric.
3. Compute I_1, I_2, I_3 for $A = \text{diag}(2, 4, 6)$.
4. Compute I_1, I_2, I_3 for $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.
5. Find the characteristic polynomial of $A = \text{diag}(a, b, c)$.
6. Find the deviatoric part of $A = \text{diag}(5, 2, 2)$.
7. Find the hydrostatic and deviatoric parts of $\sigma = \text{diag}(120, 60, 30)$.
8. Prove that the deviatoric tensor has zero trace.

9. Prove that $x_i W_{ij} x_j = 0$ for skew-symmetric W .
10. Show that $x_i A_{ij} x_j$ depends only on the symmetric part of A .
11. For $\mathbf{v} = (-\omega y, \omega x, 0)$, compute the velocity gradient and split it into symmetric and skew parts.
12. For $\mathbf{v} = (ax, by, cz)$, compute the invariants of $\nabla \mathbf{v}$.

7.6 Part V: Covariant and Contravariant Components

Final Practice V: Metrics, Raising and Lowering

1. Explain the difference between $\mathbf{g}_i$ and $\mathbf{g}^i$.
2. Prove $\mathbf{g}^i \cdot \mathbf{g}_j = \delta_j^i$ for an orthogonal coordinate system.
3. In cylindrical coordinates, compute $\mathbf{g}_\rho, \mathbf{g}_\phi, \mathbf{g}_z$.
4. In cylindrical coordinates, compute $\mathbf{g}^\rho, \mathbf{g}^\phi, \mathbf{g}^z$.
5. In spherical coordinates, compute $\mathbf{g}_r, \mathbf{g}_\theta, \mathbf{g}_\phi$.
6. In spherical coordinates, compute $\mathbf{g}^r, \mathbf{g}^\theta, \mathbf{g}^\phi$.
7. Given cylindrical physical components $(A_\rho, A_\phi, A_z) = (3, 4, 5)$, find A^i and A_i .
8. Given spherical physical components $(A_r, A_\theta, A_\phi) = (2, 3, 4)$, find A^i and A_i .
9. Prove $\mathbf{A} \cdot \mathbf{B} = g_{ij} A^i B^j$.
10. Derive $\nabla^2 f = \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} g^{ij} \partial_j f)$ for the cylindrical case.

7.7 Part VI: Complex Numbers and Hyperbolic Functions

Final Practice VI: Complex and Hyperbolic Problems

1. Write $z = -1 + i$ in polar and exponential form.
2. Write $z = -\sqrt{3} - i$ in polar and exponential form.
3. Compute $(\sqrt{3} + i)^6$ using polar form.
4. Find all roots of $z^4 = 16$.
5. Find all roots of $z^3 = -8$.
6. Solve $z^5 = 32e^{i\pi/2}$.
7. Prove Euler's formula from the power series of e^x , $\sin x$, and $\cos x$.
8. Derive De Moivre's theorem.
9. Express $\cos(x + iy)$ in real and imaginary parts.
10. Express $\sin(x + iy)$ in real and imaginary parts.
11. Prove $\cos(ix) = \cosh x$.
12. Prove $\sin(ix) = i \sinh x$.

13. Prove $\tanh(ix) = i \tan x$.
14. Derive $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$.
15. Derive $\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$.

7.8 Part VII: Mixed Challenge Problems

Final Practice VII: Mixed Challenge Set

1. A scalar field is given by $T = 1/r$. Compute ∇T , $\nabla^2 T$ for $r \neq 0$, and interpret the singularity at the origin qualitatively.
2. A velocity field is $\mathbf{v} = a\rho\hat{\mathbf{e}}_\rho + bz\hat{\mathbf{k}}$ in cylindrical coordinates. Find $\nabla \cdot \mathbf{v}$. For what relation between a and b is the flow incompressible?
3. A purely azimuthal field is $\mathbf{A} = A_\phi(r, \theta)\hat{\mathbf{e}}_\phi$ in spherical coordinates. Write the general formula for its curl.
4. A tensor A has eigenvalues 1, 2, 5. Find I_1, I_2, I_3 and write its characteristic polynomial.
5. For $A_{ij} = x_i x_j$, compute $\text{tr}(A)$, $A_{ij}A_{ij}$, and $\partial_j A_{ij}$.
6. Use $\varepsilon_{ijk}\varepsilon_{ilm}$ to derive Lagrange's identity $|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2|\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2$.
7. Show that $\nabla \cdot (f\mathbf{A} \times \mathbf{B})$ can be expanded using product identities.
8. In spherical coordinates, compute ∇f and $\nabla^2 f$ for $f = f(r)$, a radial-only scalar field.
9. In cylindrical coordinates, compute ∇f and $\nabla^2 f$ for $f = f(\rho)$, an axisymmetric radial field.
10. Explain why a formula that is true in Cartesian components may need correction terms in curvilinear coordinates if components are not physical components.
11. A complex number is multiplied by $2e^{i\pi/3}$. Describe the geometric action.
12. Solve $e^{2z} = 1$ for complex z .
13. Show that $\cosh x$ and $\sinh x$ parametrise the unit hyperbola $X^2 - Y^2 = 1$.
14. Compare $\cos^2 x + \sin^2 x = 1$ with $\cosh^2 x - \sinh^2 x = 1$. Explain the sign difference using exponentials.
15. Build a one-page personal formula sheet from memory covering all topics in this workbook.